

# Bazaar Bounty

**"The Fight for the Fruitful Fortune!"**



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# Contents

|          |                                  |          |
|----------|----------------------------------|----------|
| <b>1</b> | <b>Formal Project Proposal</b>   | <b>1</b> |
| 1.1      | Game Description . . . . .       | 1        |
| 1.1.1    | Overview . . . . .               | 1        |
| 1.1.2    | Background Story . . . . .       | 2        |
| 1.1.3    | Design Decisions . . . . .       | 2        |
| 1.2      | "Big Idea" Bullseye . . . . .    | 5        |
| 1.3      | Technical Achievement . . . . .  | 6        |
| 1.4      | Development Schedule . . . . .   | 6        |
| 1.4.1    | Layered Task Breakdown . . . . . | 6        |
| 1.4.2    | Task List . . . . .              | 8        |
| 1.4.3    | Timeline . . . . .               | 9        |
| 1.5      | Assessment . . . . .             | 9        |
| 1.6      | Contributions . . . . .          | 10       |

# Formal Project Proposal

## 1.1 Game Description

|                       |                                     |
|-----------------------|-------------------------------------|
| <b>Name</b>           | Bazaar Bounty                       |
| <b>Tagline</b>        | The Fight for the Fruitful Fortune! |
| <b>Genre</b>          | Rogue-like Dungeon Crawler          |
| <b>Perspective</b>    | Third-person Top-down               |
| <b>Nr. of players</b> | Single Player                       |
| <b>Input method</b>   | Xbox Controller                     |

### 1.1.1 Overview

*Bazaar Bounty* is a 2D rogue-like game with a top-down camera view. The protagonist is a mercenary fighting to loot vegetables and fruits to help his/her neighbors survive the severe famine. As a result, players' goal is to clear as many stages (maps filled with hostile forces) as possible while collecting more vegetables and fruits in the process. Vegetables and fruits are also consumable items providing various buffs to players, which is a crucial game mechanic in game. The game is a single-player experience, and the input method is through an Xbox controller. The main feature of the game is introducing a duel wielding weapon combat mechanic into our rogue-like game. In the meanwhile, we also aim to implement generative map generation and intelligent enemy AI. These technical features are expected to enrich our game content, adding more depth and flavor to the overall gaming experience.

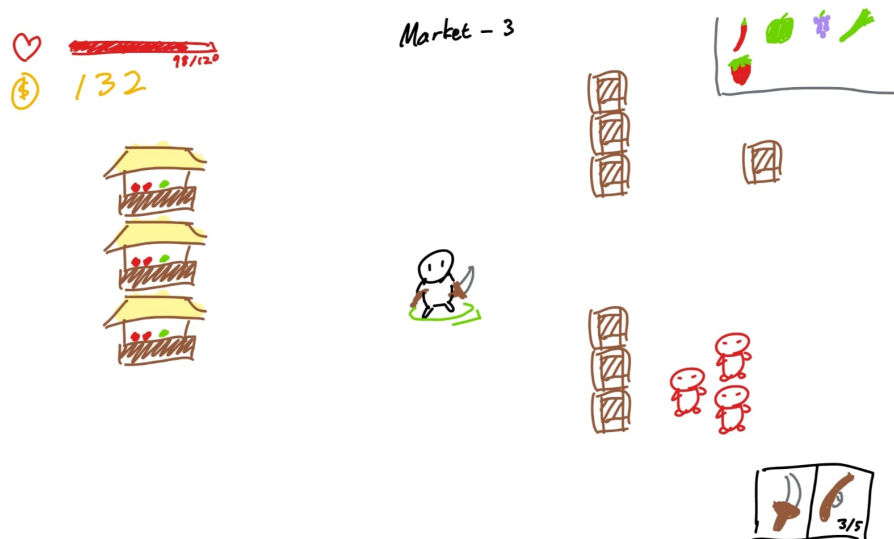
## 1.1.2 Background Story

The story begins in the late Middle Ages. A severe famine raged across the land and caused tens of thousands of deaths. The protagonist is a mercenary in a small town. To save the town from the deadly famine, he/she decided to fight for the neighbors to collect as much food as possible. The protagonist try to rob vegetables and fruits from bandits, enemy mercenaries and towns, then he sold them to villagers for money. As a result, the protagonist becomes a merchant and opens a grocery store in hometown, and the food comes mainly from his loot from enemies. Moreover, there are more than just ordinary vegetables and fruits in this land - many of them actually contain secret powers that can power up those who consume them. Playing as the mercenary hero, the player must collect magical fruits and veggies to help them on their journey.

## 1.1.3 Design Decisions

### Map Design

In *Bazaar Bounty* players play as the mercenary hero to fight in various maps. From bandit camps, to marketplaces in hostile nations, to mysterious dungeons. As a roguelike game, these maps are randomly generated with certain rules and the difficulty increases as the player progresses to deeper maps. In order to sell fruits and vegetables to the villagers, certain checkpoints would appear on some maps where players could sell food and buy equipment.



**Figure 1.1:** A Simple Mocked-up Screenshot of the Game

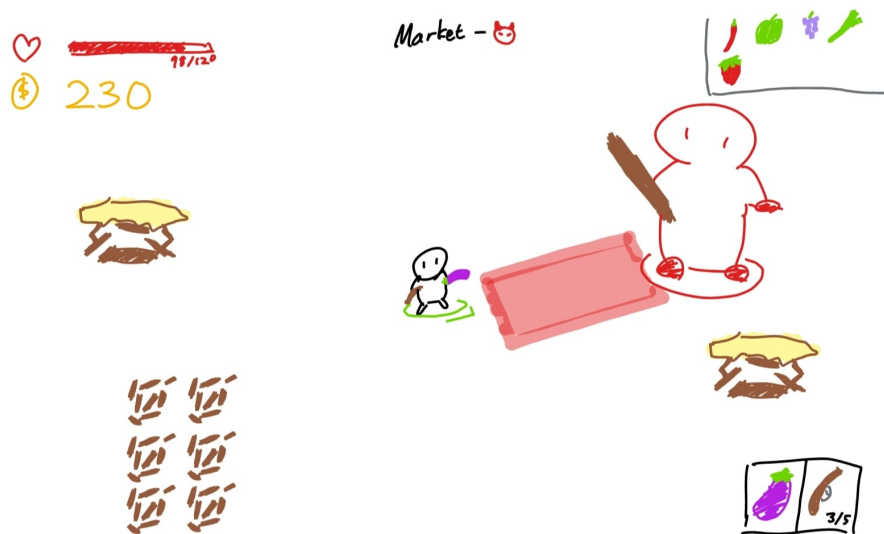
### Vegetables and Fruits

Fruits and vegetables play an important role in the game. Not only are they food that players can sell to the villagers, but they are also buff items that are used to power up players so that they

can continue on their journey. It is up to the player to decide whether to eat them immediately to get the buff or sell them for money to buy other items.



**Figure 1.2:** A Shop appeared in a Map and Route Selection



**Figure 1.3:** Encountering Enemies

## Dual-Wielding Combat Mechanic

The game features a dual-wielding combat mechanic - a primary melee weapon on the main hand and a ranged weapon on the off-hand for support. We took this combat system from action RPGs like *Dark Souls* or *Bloodborne* and integrated it into our roguelike game. This combat system can reflect the background setting that the protagonist is an experienced medieval mercenary expert various weapons, and provide players options for their favorite combat style. The desired input method for this would be the use of Xbox controllers. In addition to

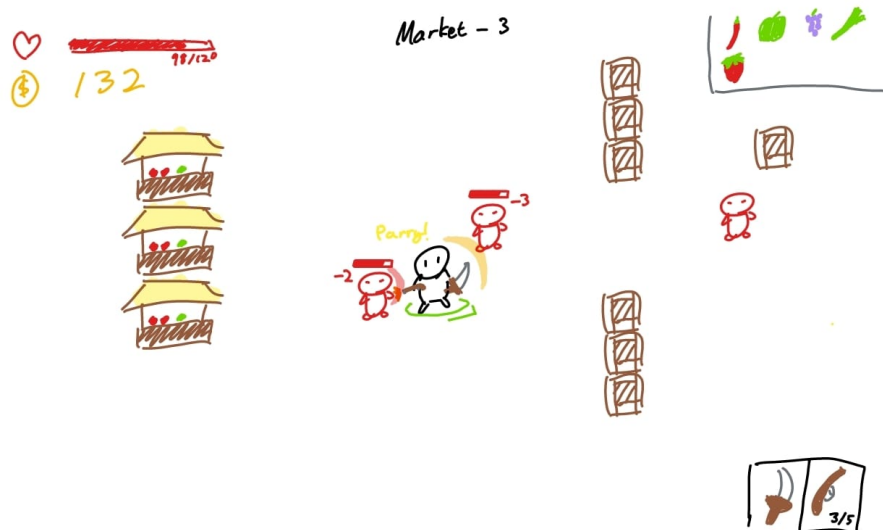
## 1 Formal Project Proposal

basic inputs such as "interact", "dash", "attack with primary/ranged weapon", there would also be special commands such as "parry" and "two-handing primary weapon".

Players always start with basically the same layout with some customization options. For example, players can choose a long sword, a dagger, or a war hammer as their primary weapon to start the game. All of these starter weapons should be well balanced so that they deal similar damage per second (DPS) to enemies, and each of them has pros and cons. A dagger might swing faster but deal less damage with each strike, while a warhammer might deal tons of damage with each strike but attack very slowly.



**Figure 1.4:** Two-handing Primary Weapon by Pressing X



**Figure 1.5:** Parrying Enemy Attack

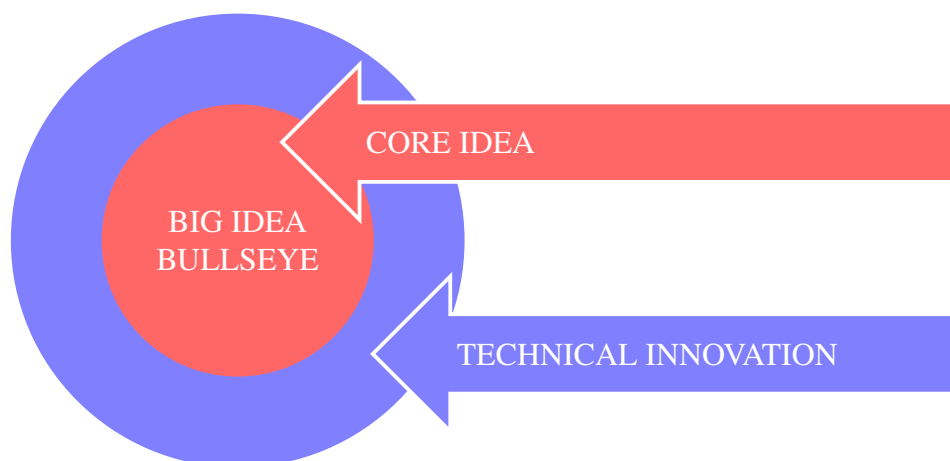
## Visual Art

The visuals of the entire game would be presented in a pixel art style. It would be quite manageable to implement such an art style in time and still manage to create appropriate assets for our game.

## 1.2 "Big Idea" Bullseye

The most important concept in an Action RPG is the combat system. In our Bazaar Bounty game introduces a dual-wielding mechanic, offering players the flexibility and also challenge of the two-handed combat. Players will wield a primary melee weapon for close-quarters combat, complemented by a secondary weapon that can either extend their reach through ranged attacks or enhance their defensive capabilities with parrying mechanics inspired by soulslike games, which provides players with freedom of movement and defense. The two-handed combat system not only diversifies combat strategies but also challenges players to adopt different strategies in various combat scenarios.

The core technical component in our game is the procedural map generation and the enemy AI. Procedural map generation ensures that each playthrough presents a unique layout of the bazaar, enriching the freshness of the exploration. A sophisticated enemy AI allows enemies to take action according to different strategies, providing an engaging combat experience. Details of technical component is described in Technical Component part.



**Figure 1.6:** Big Idea Bullseye

## 1.3 Technical Achievement

### Procedural Map Generation

The first core technical component of our game involves implementing a procedural map generation system. This method guarantees that every new game or level introduces a completely new map. It can enhance the replayability and maintain a fresh experience for players.

For procedural map generation, we have to ensure the randomness of the map as well as the validity of the map, which means there must exist a clear path that connects each region. Perlin noise can be used to generate cave-like map with randomness. Meanwhile, Cellular Automata algorithms are applied to refine these landscapes, ensuring all areas are interconnected and the game remains playable.

### Enemy AI

In a single-player action RPG, having a robust enemy AI system is essential. The enemy AI defines enemies' behavior such movement, attacking or interacting with the environment by using a combination of pathfinding algorithm and behavior tree systems.

For navigation and movement, enemies rely on the A\* algorithm. This allows them to move intelligently towards the player, planning their attacks. The behaviour of enemies can be defined by a behavior tree, which is a hierarchical model of actions and decisions that allows for complex behaviors. Enemies can interact with the environment and employ various strategies such as hiding, using shields, or quick dashes, depending on their level and type. These strategies can be built on a generic behavior tree, designed to make the game more challenging and unique.

## 1.4 Development Schedule

### 1.4.1 Layered Task Breakdown

#### Functional Minimum

The foundation of the game is established by implementing one room containing a singular enemy equipped with rudimentary AI. The player should be able navigate the room and defeat the enemy with a singular melee weapon.

#### Low Target

The game is expanded to have branching one-way paths containing various room structures. A couple enemies with differentiated AIs are introduced. The player can choose to use a right-handed melee weapon or a left-handed ranged weapon to engage in combat. Some artistic coherence is established to ensure visual consistency.

### **Desired Target**

Our desired state of the game relies on polished hybrid combat. The player can choose between dual wielding melee & ranged or two handed melee only. The ranged weapon can additionally be used to parry and block incoming projectiles. Various power-ups themed after fruit and vegetables can be acquired during a run.

### **High Target**

A singular boss enemy with various attacks and elaborate AI is added. Throughout a run a currency can be collected to be traded against vendor upgrades and items. Additional status effects, as well as sources of acquiring and losing said status effects are implemented. A run progression map signifies the player's progress.

### **Extras**

Additional bosses and enemy types, that vary in theme and functionality can be implemented. Meta progression allows the players to have long-term progression goals.

## 1.4.2 Task List

| Task | Who                      | Description                           | Hours |
|------|--------------------------|---------------------------------------|-------|
| 1    | All                      | Brainstorm design                     | 3.5   |
| 2    | Ömer, Willy, Kang, Zhang | Formal game proposal report           | 2.5   |
| 3    | Liu                      | Formal game proposal slides           | 2.5   |
| 4    | All                      | MonoGame basics                       | 10    |
| 5    | All                      | Physical game prototype               | 5     |
| 6    | Ömer, Willy, Kang        | Basic player & enemy implementation   | 50    |
| 7    | Zhang, Liu               | Basic room implementation             | 50    |
| 8    | All                      | First playable demo video             | 2     |
| 9    | Ömer                     | Ranged weapon addition                | 30    |
| 10   | Willy, Kang              | Enemy addition                        | 30    |
| 11   | Zhang, Liu               | Procedural room generation            | 30    |
| 12   | All                      | Interim report and video              | 5     |
| 13   | Ömer, Willy              | Two handing melee weapon and parrying | 20    |
| 14   | Kang, Zhang, Liu         | Power-ups                             | 20    |
| 15   | All                      | Alpha release report and video        | 5     |
| 16   | All                      | Playtests report and slides           | 5     |
| 17   | All                      | Conclusions report                    | 5     |
| 18   | All                      | Final presentation slides             | 5     |

**Table 1.1:** Task List

### 1.4.3 Timeline

| Task | Who        | Week |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
|------|------------|------|---|---|---|---|---|---|---|---|----|----|----|----|----|----|
|      |            | 1    | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 1    | All        |      | X |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 2    | Ö, W, K, Z |      | X |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 3    | L          |      | X |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 4    | All        |      |   | X |   |   |   |   |   |   |    |    |    |    |    |    |
| 5    | All        |      |   | X |   |   |   |   |   |   |    |    |    |    |    |    |
| 6    | Ö, W, K    |      |   |   | X | X | X |   |   |   |    |    |    |    |    |    |
| 7    | Z, L       |      |   |   | X | X | X |   |   |   |    |    |    |    |    |    |
| 8    | All        |      |   |   |   | X |   |   |   |   |    |    |    |    |    |    |
| 9    | Ö          |      |   |   |   |   |   | X | X |   |    |    |    |    |    |    |
| 10   | W, K       |      |   |   |   |   |   | X | X |   |    |    |    |    |    |    |
| 11   | Z, L       |      |   |   |   |   |   | X | X |   |    |    |    |    |    |    |
| 12   | All        |      |   |   |   |   |   |   | X |   |    |    |    |    |    |    |
| 13   | Ö, W       |      |   |   |   |   |   |   |   | X | X  |    |    |    |    |    |
| 14   | K, Z, L    |      |   |   |   |   |   |   |   | X | X  |    |    |    |    |    |
| 15   | All        |      |   |   |   |   |   |   |   |   | X  |    |    |    |    |    |
| 16   | All        |      |   |   |   |   |   |   |   |   |    | X  |    |    |    |    |
| 17   | All        |      |   |   |   |   |   |   |   |   |    |    | X  |    |    |    |
| 18   | All        |      |   |   |   |   |   |   |   |   |    |    |    | X  | X  |    |

**Table 1.2:** Timeline

## 1.5 Assessment

The main strength of the game, outside of the typical advantages of high replay value that comes inherently with the roguelike genre, will be its fluid and dynamic combat system inspired by the Dark Souls series. The ability to freely swap between dual-wielding or two-handing will add complexity to the combat that the players can explore to their liking. The high variety of weapons and powerups will also prevent the combat system from feeling repetitive, as players can mix and match as they progress through the game.

## 1 Formal Project Proposal

The game builds a world in the Middle Ages, where the protagonist is a mercenary, fighting bandits and other enemies for their loot. The player will feel compelled to move through the game's progression, as what they're doing is considered a good deed where they punish the evil and help the weak. In addition, the fruits and vegetables offering special powerups add a touch of fantasy to the setting. This combined with the combat system should appeal to RPG players and the main target audience of roguelike fans.

The main criteria to judge the success of a design would be player reception towards the combat system. In an ideal case, the combat system should feel intuitive, fun and exciting, while introducing a good depth to be explored by the dual-wielding and parry system. Secondly, the player should be able to understand the purpose of the protagonist fighting through the bandits. Should the protagonist's motivation feel arbitrary, it may break the player's immersion in the virtual world the game builds.

### 1.6 Contributions

|                       |                                                                              |
|-----------------------|------------------------------------------------------------------------------|
| <b>Alkaya Ömer</b>    | Brainstormed design and contributed to report.                               |
| <b>Willy Shih</b>     | Brainstormed design and contributed to report.                               |
| <b>Yingzhe Liu</b>    | Brainstormed design and contributed to slides.                               |
| <b>Minsung Kang</b>   | Brainstormed design, worked on mockup screenshots and contributed to report. |
| <b>Shenghao Zhang</b> | Brainstormed design and contributed to report.                               |